

Practice problems and exercises on Linear Algebra

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Information

- There are two sections
 - Section A contains Problems along with solutions.
 - Section B contains exercises for you to try and solve.
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Section A: Practice problems with solutions

Q1. Let set

$$S = \left\{ \begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix}, \begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix} \right\}$$

Check whether S is a linearly independent set.

Solution to Q1

By the definition of linear independence, if linear combination of the vectors of the set equated to zero vector implies that the coefficients are all equal to zero uniquely, then the set is linearly independent.

$$\alpha_1 \begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix} + \alpha_2 \begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\implies \begin{pmatrix} \alpha_1 \\ \frac{\alpha_1}{2} \end{pmatrix} + \begin{pmatrix} \frac{\alpha_2}{2} \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\implies \begin{pmatrix} \alpha_1 + \frac{\alpha_2}{2} \\ \frac{\alpha_1}{2} + \alpha_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\implies \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Check that

$$\det \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix} = \frac{3}{4} \neq 0,$$

so

$$\begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix}$$

is invertible. Hence,

$$\implies \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix}^{-1} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

So, S is linearly independent.

Q2. Let set

$$S = \left\{ \begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix}, \begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$

Check whether S is a linearly independent set.

Solution to Q2

By the definition of linear independence, if a linear combination of the vectors of the set equated to the zero vector implies that the coefficients are all equal to zero uniquely, then the set is linearly independent.

$$\begin{aligned}\alpha_1 \begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix} + \alpha_2 \begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 0 \\ 1 \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ \Rightarrow \begin{pmatrix} \alpha_1 \\ \frac{\alpha_1}{2} \end{pmatrix} + \begin{pmatrix} \frac{\alpha_2}{2} \\ \alpha_2 \end{pmatrix} + \begin{pmatrix} 0 \\ \alpha_3 \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ \Rightarrow \begin{pmatrix} \alpha_1 + \frac{\alpha_2}{2} \\ \frac{\alpha_1}{2} + \alpha_2 + \alpha_3 \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ \Rightarrow \begin{pmatrix} 1 & \frac{1}{2} & 0 \\ \frac{1}{2} & 1 & 1 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix}\end{aligned}$$

After substitution, one will find that

$$\begin{pmatrix} \alpha_1 \\ -2\alpha_1 \\ \frac{3\alpha_1}{2} \end{pmatrix}$$

is a solution. Hence, there are non-zero solutions. Therefore, S is linearly dependent.

Q3. Let $\mathbf{a}$ and $\mathbf{b}$ be two vectors given by

$$\mathbf{a} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

- (a) Find the projection of $\mathbf{a}$ onto $\mathbf{b}$.
- (b) Find an orthonormal vector $\mathbf{v}$ orthogonal to $\mathbf{b}$.
- (c) Check whether $\{\mathbf{v}, \mathbf{b}\}$ is a basis for $\mathbb{R}^2$.
- (d) Express $\mathbf{a}$ as a linear combination of $\mathbf{v}$ and $\mathbf{b}$.

Solution to Q3

$$\text{Given: } \mathbf{a} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Projection of $\mathbf{a}$ onto $\mathbf{b}$

The projection of $\mathbf{a}$ onto $\mathbf{b}$ is given by

$$\text{proj}_{\mathbf{b}} \mathbf{a} = \frac{\mathbf{a}^T \mathbf{b}}{\mathbf{b}^T \mathbf{b}} \mathbf{b}$$

$$\mathbf{a}^T \mathbf{b} = (2, 1) \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 4, \quad \mathbf{b}^T \mathbf{b} = (1, 2) \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 5$$

$$\text{proj}_{\mathbf{b}} \mathbf{a} = \frac{4}{5} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} \frac{4}{5} \\ \frac{8}{5} \end{pmatrix}$$

Finding an orthonormal vector $\mathbf{v}$ orthogonal to $\mathbf{b}$

A vector orthogonal to $\mathbf{b} = (1, 2)^T$ is

$$\mathbf{u} = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \quad \text{since } \mathbf{b}^T \mathbf{u} = 0$$

The norm of $\mathbf{u}$ is

$$\|\mathbf{u}\| = \sqrt{2^2 + (-1)^2} = \sqrt{5}$$

Hence the orthonormal vector is

$$\mathbf{v} = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

Checking whether $\{\mathbf{v}, \mathbf{b}\}$ is a basis for $\mathbb{R}^2$

Consider a linear combination

$$\alpha \mathbf{v} + \beta \mathbf{b} = \mathbf{0}.$$

Taking dot product of both sides with $\mathbf{v}$, we get

$$\alpha \mathbf{v}^T \mathbf{v} + \beta \mathbf{b}^T \mathbf{v} = 0.$$

Since $\mathbf{v}$ is orthonormal and orthogonal to $\mathbf{b}$,

$$\mathbf{v}^T \mathbf{v} = 1, \quad \mathbf{b}^T \mathbf{v} = 0.$$

Hence,

$$\alpha = 0.$$

Substituting $\alpha = 0$ in the original equation,

$$\beta \mathbf{b} = \mathbf{0}.$$

Since $\mathbf{b} \neq \mathbf{0}$, it follows that

$$\beta = 0.$$

Therefore, the vectors $\mathbf{v}$ and $\mathbf{b}$ are linearly independent, and hence $\{\mathbf{v}, \mathbf{b}\}$ is a basis for $\mathbb{R}^2$.

Expressing vector $\mathbf{a}$ as a linear combination of $\mathbf{v}$ and $\mathbf{b}$

Since $\{\mathbf{v}, \mathbf{b}\}$ is an orthogonal basis,

$$\mathbf{a} = (\mathbf{a}^\top \mathbf{v})\mathbf{v} + \text{proj}_{\mathbf{b}} \mathbf{a}$$

$$\mathbf{a}^\top \mathbf{v} = (2, 1) \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \frac{3}{\sqrt{5}}$$

$$\mathbf{a} = \frac{3}{\sqrt{5}}\mathbf{v} + \frac{4}{5}\mathbf{b}$$

$$\boxed{\mathbf{a} = \frac{3}{\sqrt{5}}\mathbf{v} + \frac{4}{5}\mathbf{b}}$$

Section B: Exercises

The following problems are taken from *Higher Algebra*, S. K. Mapa, 12th Edition.

1. Show that the set S of vectors is linearly independent in $\mathbb{R}^3$.

(i) $S = \{(1, 2, 3), (2, 3, 1), (3, 1, 2)\}$

(ii) $S = \{(0, 1, 1), (1, 0, 1), (1, 1, 0)\}$

2. Show that the set S of vectors is linearly independent in $\mathbb{R}^4$.

(i) $S = \{(1, 2, 3, 0), (2, 3, 0, 1), (3, 0, 1, 2)\}$

(ii) $S = \{(1, 1, 1, 0), (1, 1, 0, 1), (1, 0, 1, 1), (0, 1, 1, 1)\}$

3. Find a basis for the row space of the following matrices

(i) $\begin{pmatrix} 1 & 0 & 3 & 2 \\ 4 & 0 & 2 & 2 \\ 0 & 3 & 1 & 4 \end{pmatrix}$.

(ii) $\begin{pmatrix} 2 & 1 & 3 & 5 \\ 3 & 4 & 1 & 2 \\ 0 & 3 & 1 & 1 \\ 5 & 5 & 4 & 7 \end{pmatrix}$.